

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA

DIPLOMA EXAMINATION (C-24)

C-24-OCT-2025

SEMESTER III , MID-II EXAM

AI/BM/CCB/CPS/CS/EC/EE/EI/ES/EV

EC-302

Digital Electronics



PCODE
243034

Exam Date: 16-10-2025

Duration: 1 Hour [01:30 PM To 02:30 PM]

Session: AN

[Total Marks: 20]

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

4 X 1 = 4

1. How many inputs a full subtractor will have?
2. Define a combinational logic circuit.
3. Draw the logic symbol of 4x1 multiplexer
4. List applications of encoder

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

2 X 3 = 6

- 5(a). Draw a Half adder circuit using NOR gates only.
----- OR -----
- 5(b). Write any 3 differences between serial adder and parallel adder.
- 6(a). Write the IC numbers of 4x1 MUX and 1x4 DEMUX, 3x8 decoder and 8x3 encoder
----- OR -----
- 6(b). Write any 3 applications for of Multiplexer.

PART-C

Instructions:

1. Answer the following questions.
2. Each question carries **FIVE** marks.

2 X 5 = 10

- 7(a). Explain the working of serial adder circuit.

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----- OR -----

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7(b). Realize a full adder using two half adders and an OR gate.

8(a). Explain the 4X1 multiplexer with logic diagram.

----- OR -----

8(b). Draw The circuit of 1-bit digital comparator and explain its operation

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